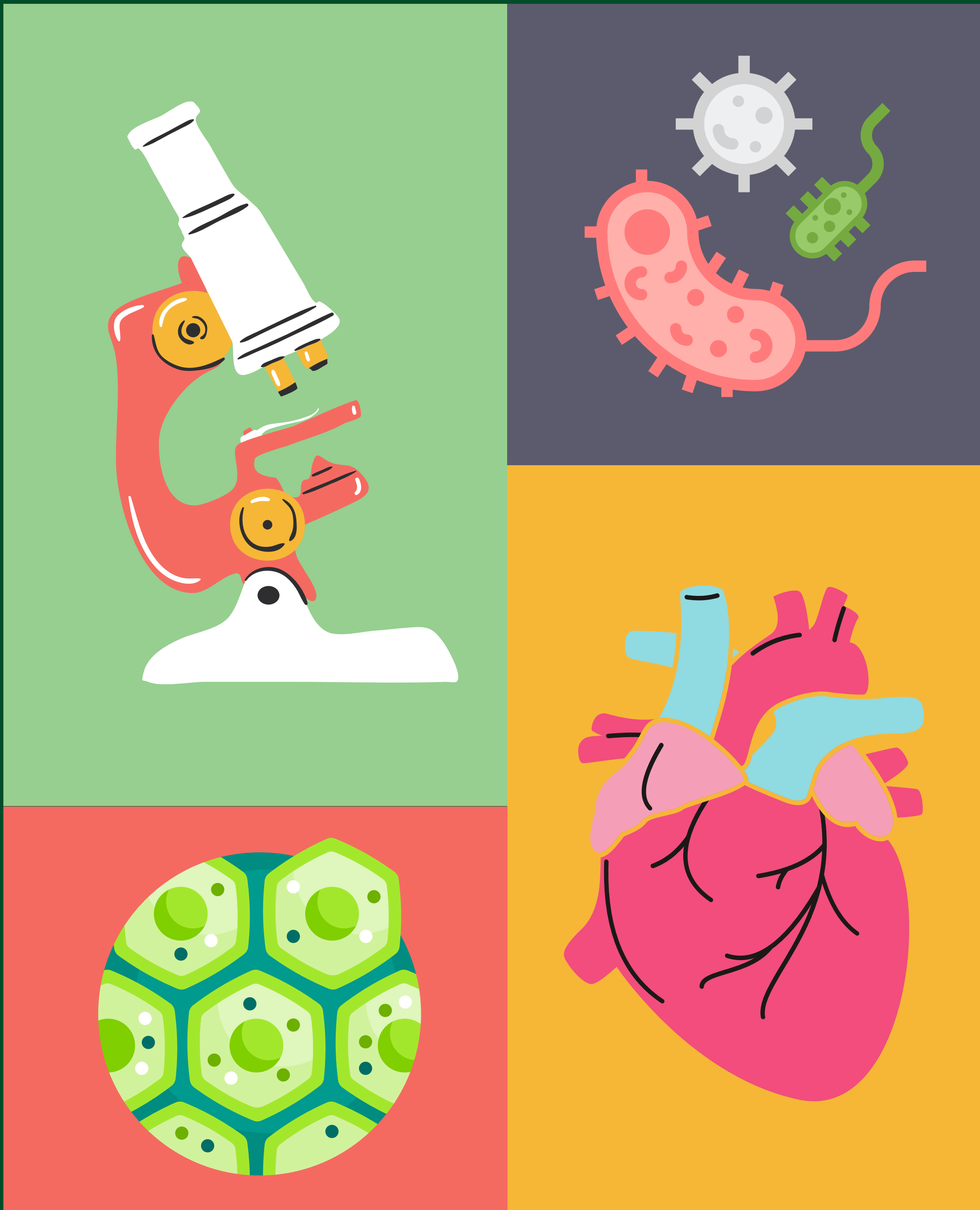


CLASS 10 NOTES

SCIENCE

Our Environment

PRASHANT KIRAD

ENVIRONMENT- It refers to the complete range of physical and biological conditions in which org. like and interact with biotic and abiotic factors.

ECOSYSTEM: All the interacting organisms in an area together with the non-living constituents of the environment form an ecosystem. E.g. forest, pond etc.

There are two types of ecosystems.

1.Natural ecosystem

2.Artificia ecosystem

- The ecosystem which exist in nature on its own. e.g., forest, lake, ocean.
- Man-made ecosystem are called artificial ecosystem. e.g., crop field, aquarium, garden.

note- Agro-ecosystem is the largest man-made ecosystem.
other examples are aquariums, botanical gardens, parks, field crops, etc.

Components of an Ecosystem

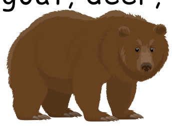
- Abiotic components (air, water, land)
- Biotic components (Plant, animals)
  - producer
  - Consumer
    - Herbivore
    - Carnivore
    - Omnivore
    - Parasites
  - Decomposers

E.M.A

1. **Biotic component:** It includes three types of organisms :

(a) Producers: All green plants, blue green algae can produce their food (Sugar and starch) from inorganic substance using light energy (Photosynthesis). Therefore, all green plants are called producers. They are also called autotrophs.

(b) Consumers: They are organisms which consume other organisms or their products as their food. All animals belong to this category. The consumers depend upon producers for their food directly or indirectly. They get their food by eating other organisms or their products. For example, man, goat, deer, fish, lion, cow, buffalo, etc., are common consumers.



The consumers can be classified into the following types :

Herbivores, carnivores, omnivores and parasites are the various type of consumers.

1. Herbivores: These are organisms (animals) which get their food by eating the producers (or plant) directly. Herbivores are also called first order consumers. Some common examples of herbivores are : deer, rabbit, rat, squirrel, goat, cattle, etc.

2. Carnivores: These are organisms (animals) which consume other animals. Therefore, carnivores feed on the flesh of herbivores. These are also called primary carnivores or second order consumers. Some common examples are snake, wild cat, jackal, frog, some birds, fishes, etc.

3. Omnivores: The organisms which feed on both plants and animals are called omnivores. Human beings are common example of omnivores because they eat both plants (For example; pulses, grams, oilseeds, fruit, etc.) and animal products (milk, meat, egg, etc.).

4.parasites - Those who live on body of host and take food from it without killing them. Egdice, cascuta etc

(c) Decomposers: They are those micro-organisms that obtain energy from the chemical breakdown of dead organisms or animals or plant wastes. These micro organisms are decomposers as they breakdown the complex organic substances into simple inorganic substances that go into the soil and are used up once more by the plants.

2. Abiotic Components: All the nonliving components such as air, water, land , CO_2 , O_2 , light etc . form abiotic .These component are physical factors such as light , temp. , water etc.

Climatic factors: These are sunlight temperature, pressure humidity, moisture, rainfall, etc. these factors affect the distribution of the organisms.

- **Light energy** (sunlight) is the primary source of energy in nearly all ecosystems. It is used by green plants (which contain chlorophyll). During photosynthesis plants manufacture organic substances by combining inorganic substances.

- **Temperature** The distribution of plants and animals is greatly influenced by extremes in temperature. The pattern of rain also affects the growth of the plant. This plant growth determines the overall variety of animals living in that place.

- **Atmospheric Gases** Oxygen is required for respiration and carbon dioxide for photosynthesis. Nitrogen is made available to plants by certain bacteria and through the action of lightening.

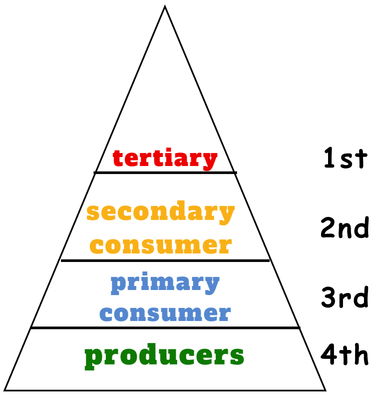
- **Wind** It helps in pollination and seed dispersal of some plants. It can remove and redistribute top soil, especially where vegetation has been reduced.

✷**Water** It is essential for life. Plant and animal habitats vary from entirely aquatic environments to very dry deserts.

E.M.A

Trophic levels:

Trophic levels are the various steps or levels in the food chain where transfer of food or energy takes place. Producers are the first trophic level, herbivores are second trophic level, carnivores or secondary consumers are third trophic level and large carnivores or tertiary consumers are the fourth trophic level.



TROPHIC LEVELS IN AN ECO-SYSTEM

E.M.A

Food chain:

Food chain is sequence of organisms through which energy is transferred in the form of food by the process of one organism consuming the other.

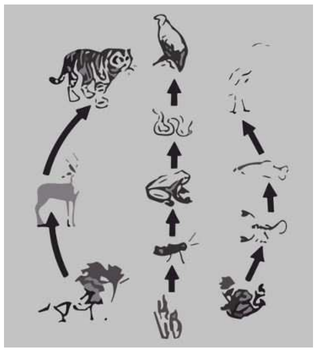
e.g: **Grass** (Producer) > **Grasshopper** (Herbivore) > **Frog** (Carnivore) > **Snake** (Carnivore) > **Eagle** (Top Carnivore)

On the basis of choice of habitat, food chains are of two types:

(1). Terrestrial food chain It is the food chain present on land.
e.g. Grass > Insects > Snake + Hawk.

(2). Aquatic food chain It is the food chain in different water bodies.

e.g. Phytoplankton — Zooplankton — Fish — shark.



Significance of Food Chain:

1.Exploring food chains helps us see how different creatures in an ecosystem share food and interact. Think of food chains as nature's delivery system, moving energy and resources between the different living parts of an ecosystem.
2.Food chains act like energy highways, carrying vital energy and resources between the different living parts of an ecosystem or even across the whole biosphere.
3.Food chains are like the engines of an ecosystem or the biosphere, making everything alive and dynamic by showing how energy flows between different creatures.
4. When harmful substances like pesticides and weedkillers travel through food chains, they can become increasingly dangerous. This can lead to more serious effects on top-level creatures, including humans. It's a reminder of how important it is to understand and manage these harmful chemicals in ecosystems.

#SECRET QUESTIONS

1.a. What is an ecosystem? List its two main components.
b. We do not clean ponds or lakes, but an aquarium needs to be cleaned regularly. Explain.

a. A self-sustaining functional unit consisting of living (biotic) and nonliving (abiotic) components, is called an ecosystem.
1. Biotic components: Plants and animals.
2. Abiotic components: Light, soil, temperature, humidity, wind, air, etc.

b. An aquarium is an artificial and incomplete ecosystem in contrast to a pond or lake which is natural, self- sustaining and complete ecosystem. In natural ecosystem, decomposers help in recycling waste. So, an aquarium needs to be cleaned regularly.

2.Mention the differences between food habits of organisms belonging to the first and third trophic level. Give one example of each of them

First Trophic Leve	Third Trophic Level
The organisms of this trophic level are plants and are also called producers.	The organisms of this trophic level are animals and are also called secondary.
They transform solar energy into chemical energy by photosynthesis. e.g., grass (all green plants)	They obtain chemical energy by eating other animals. e.g., all carnivores (like lion).

3.State one important function of ozone layer in the atmosphere. How is it formed there? Which compounds are responsible for the depletion of ozone layer? How do these compounds enter into the atmosphere?

Ozone present in the upper regions of the atmosphere protects us from dangerous UV radiations. Formation of ozone layer :

Ozone at the higher levels of the atmosphere is a product of UV radiations acting on oxygen (O2) molecule. The higher energy UV radiations split apart some molecular oxygen (O2) into free oxygen (O) atoms. These atoms then combine with the molecular

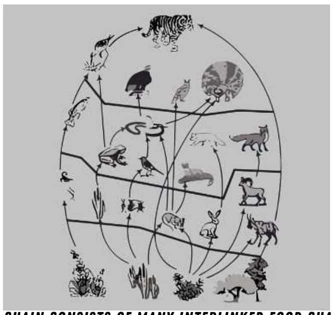
oxygen to form ozone as shown: $O_2 \xrightarrow{UV} O + O$

$$O + O_2 \longrightarrow O_3 \quad \text{Ozon}$$

E.M.A

FOOD WEB:

Food web is the network of various food chains which are interconnected at various tropic levels. Since an organism can occupy position in more than one food chain, in a food web it occupies more than one trophic level.



FOOD CHAIN CONSISTS OF MANY INTERLINKED FOOD CHAINS

E.M.A

ENERGY FLOW

Energy is accumulated by the primary producers and its transferred through food chain to different trophic This phenomenon is called energy flow. It is unidirectional and there is no recycling or going back to previous level, whenever energy is transferred from one to another, some energy is always lost.

The flow of energy in an ecosystem:

The energy in a food chain only moves in one direction. It goes from the plants (the autotrophs) to the plant-eating animals (the herbivores) and doesn't go back to the sun.

The flow of energy in different trophic levels:

- Energy in a food chain only goes in one direction. Green plants capture a small part of sunlight (about 1%) and turn it into food energy.
- According to the **10 percent law**, only about 10% of this energy is passed on to the next level of creatures. The rest, the remaining 90%, is used up by the current level of organisms for their life processes like digestion, growth, and reproduction.
- Because energy decreases at each step, most food chains consist of 3 to 4 levels of different creatures.
- Biological magnification is when harmful chemicals become more concentrated as you move up the food chain. This can be a problem because the highest concentration of these chemicals often ends up in human bodies since humans are typically at the top of many food chains.
- Ozone (O3) is a special form of oxygen made up of three oxygen atoms. Ozone plays a crucial role in protecting the Earth's surface from the sun's harmful ultraviolet radiation.

$$O_2 + O \rightarrow O_3$$

Depletion of ozone layer: Ozone layer gets depleted - due to the use of chemicals called aerosol, spray propellants like chlorofluorocarbons. Depletion of ' ozone layer would cause skin cancer in men and animals and severe damage to the plants.

Biological magnification happens when harmful, non-biodegradable chemicals like pesticides accumulate in organisms as you move up the food chain, increasing their concentration.

TOP 7 QUESTIONS

1. Distinguish between biodegradable , and nonbiodegradable substances. ' List two effects of each of them on our environment.

Biodegradable	Non-biodegradable
• Substances that are broken down by biological processes are said to be biodegradable.	• Substances that are not broken down by biological processes are said to be nonbiodegradable.
• These substances get recycled and, therefore, do not require any dumping sites.	• These substances require a lot of space for dumping which causes wastage of land.

2. a. What is the height of ozone from the equator?
   b. Name the rays against which ozone layer provides protection
   c. Name one effect of depletion of ozone.

i. 10 to 16 km.
ii. UV rays.
iii. Global wanning.

3. State two advantages of conserving
(i) forests
(ii) wild life

- Advantages of conserving forest are termed as biodiversity hotspots. They have large number of species of plants and animals.
  (a) They purify air, help in recharging groundwater, bring rains and maintain the fertility of soil.
  (b) They are also a source of income for tribal people.
- Wild life is important
  (a) To preserve bio-diversity.
  (b) As each species has a position in the food chain so wildlife helps in balancing the nature.

4. Why are green plants called 'producers' ?

'Producers' are the organisms which prepare their own food in the presence of sunlight and chlorophyll. Therefore green plants are called producers as they prepare their own food.

5. Energy flow in a food chain is unidirectional. Justify this statement. Explain how the pesticides enter a food chain and subsequently get into our body.

a. Energy moves progressively through the various trophic levels and is no longer available to the previous trophic level. The energy captured by autotrophs does not revert back to the solar input. Therefore flow of energy is unidirectional.

b. Pesticides, used for crop rotation when washed down into the soil/water body, are absorbed by the plant/producer along with water and minerals. Being non-biodegradable these chemicals get accumulated progressively in the food chain and enter our body.

6. Food web increases the stability of an ecosystem. Justify.

Food web depicts feeding connection in an ecological community. It consists of many food chains. Thus, if any of the organism becomes endangered or extinct, the one who is dependent on it has an alternative option available to him for its survival. In this way, food web increases stability in an ecosystem.

7. What is wild life? How is it important? How is it being protected by government of India?

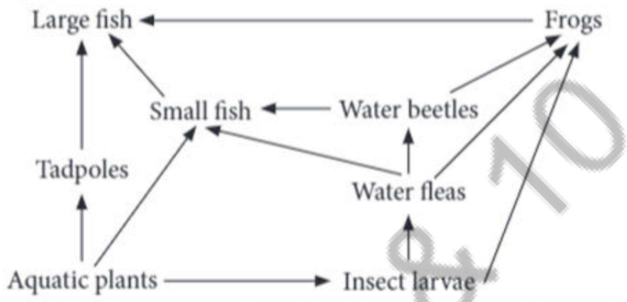
Wild life means our flora and fauna. It is important:
a. to preserve bio-diversity.
b. as each species has a position in the food chain so wildlife helps in balancing the nature.

Various species of plants and animals are preserved in botanical gardens, national parks, zoological parks and wildlife sanctuaries.

CASE BASED COUNTRY

1. Alternatives are always available in nature which results in a sort of interlocking pattern or food web. For instance, if a particular species of producers get destroyed by a disease in an ecosystem, herbivores of that area can feed on other species. Also in a food web, any given species may operate simultaneously at more than one trophic level.

(i) Refer to the given food web.



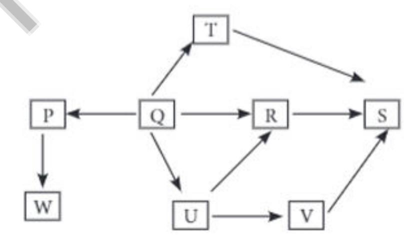
What will be the effect on the food web if population of water fleas get eliminated?

(a) Population of water beetles will increase.
(b) Population of insect larvae will remain unaffected.
(c) Population of small fish will decrease.
(d) Population of frog will increase.

(ii) In the given food web, which organism operates at both primary and tertiary consumer level?

(a) Small fish (b) Frog (c) Water fleas (d) Tadpole

(iii) Refer to the given food web.



[Letters P – W in the given food web represent living things].

Which of the following statements regarding the given food web is correct?

(a) When population of U decreases, population of R and V will increase.
(b) When population of Q decreases, population of R, T and U will increase.
(c) When population of R increases, population of S will increase.
(d) When population of S increases, population of R, T and U will also increase.

(iv) Which of the following statements is correct?

(a) Food webs provide alternative pathways of food availability.
(b) Greater alternatives available in a food web make the ecosystem more stable.
(c) Food webs also help in ecosystem development.
(d) All of these

(v) Food webs make a natural ecosystem _______ than an man-made ecosystem.

(a) unstable (b) stable (c) variable (d) inconsistent

2. Ozone layer is present in the earth's atmosphere. It is in the form of a protective shield. It contains three oxygen atoms (O3) which are formed as a consequence of photochemical reactions in the environment. Ozone absorbs harmful ultraviolet radiations of the sun. In this way, it protects all living beings on the earth. The thinning of ozone layer due to various human activities allows more UV radiations to pass through it which leads to harmful effects on man, animals and plants.

(i) Ozone layer is present in which layer of the atmosphere?
(a) Troposphere (b) Mesosphere (c) Stratosphere (d) Thermosphere

(ii) Enhanced UV-radiations would affect humans and other animals causing
(a) skin cancer
(b) blindness and increased chances of cataract in eyes
(c) malfunctioning of the immune system
(d) all of these.

(iii) Read the given statements regarding ozone.
I. Ozone hole was first discovered over Montreal in 1976.
II. Ozone is a result of photochemical reactions in which starting molecule is oxygen.
III. Harmful chemicals produce active chlorine in presence of UV radiations, that destroys ozone layers.
IV. Ozone absorbs UV-radiations in the range of 800 – 1100 Å.
Select the option that correctly identifies them as true (T) and false (F).

	I	II	III	IV
(a)	F	T	T	F
(b)	F	F	T	T
(c)	T	T	F	F
(d)	T	F	T	T

(iv) Which of the following are related to depletion of ozone layer?
(a) Chlorofluorocarbons (b) Halons
(c) Carbon tetrachloride (d) All of these

(v) Refer to the given events regarding thinning of ozone layer and arrange them in a sequence.
I. Active chlorine is produced in presence of UV radiations.
II. CFCs are released in the air.
III. Ozone layer in the stratosphere become thin.
IV. CFCs enter from troposphere into stratosphere.
V. Use of CFCs in refrigerators and air conditioners as coolants.
VI. Active chlorine destroy ozone by converting it into oxygen.
(a) V → II → I → VI → IV → III (b) V → II → IV → I → VI → III
(c) V → I → II → III → VI → IV (d) V → IV → II → I → III → VI